

4.3 Representing Functions as Power Series Worksheet

1. Find a power series representation of the function $g(x) = \frac{3x}{1-x^2}$ and state its interval of convergence.

2. Consider the series $\sum_{n=0}^{\infty} \frac{1}{3^n} x^n$. Find the function represented by this series. Determine the interval of convergence.

3. Consider the series $\sum_{n=0}^{\infty} 2^n x^n$.

- a) Find the function represented by this series. Determine the interval of convergence.

- b) Use your knowledge from part (a) to find a function which represents the series $\sum_{n=1}^{\infty} 2n(2x)^{n-1}$ and state the radius of convergence.

- c) Use your knowledge from part (b) to determine the exact sum of the series $\sum_{n=0}^{\infty} 2n\left(\frac{1}{4}\right)^{n-1}$.

4. Consider the series

$$\frac{2}{3} - \frac{10(x-1)}{9} + \frac{50(x-1)^2}{27} - \frac{250(x-1)^3}{81} + \frac{1250(x-1)^4}{243} + \dots$$

- a) For what values of x does the following series converge?

- b) What is the sum of the series for these values of x ?

5. Consider the function $f(x) = \frac{2x}{2+x^2}$.

a) Find a power series representation for this function and state its interval and radius of convergence.

b) Use information from part (a) to determine a power series representation for $g(x) = \frac{4-2x^2}{(2+x^2)^2}$ and state its radius of convergence.

6. Suppose that $\sum_{n=0}^{\infty} a_n x^n$ is a power series with interval of convergence $(-1,1)$. Find the interval of convergence for the power series $\sum_{n=0}^{\infty} a_n \left(\frac{x}{2}\right)^n$.

7. Find a power series representation for $g(x) = \frac{1}{(1-x)(x-2)}$. Find the interval of convergence for power series.

8. Consider the series

$$\frac{3}{4} - \frac{3(x-3)}{8} + \frac{3(x-3)^2}{16} - \frac{3(x-3)^3}{32} + \frac{3(x-3)^4}{64} - \frac{3(x-3)^5}{128} + \dots$$

- a) For what values of x does the following series converge?

b) What is the sum of the series for these values of x ?

9. Consider the series $\sum_{n=0}^{\infty} \frac{5}{2} x^{3n+1}$.

a) Find a function $f(x)$ which is represented by this series and state the interval of convergence.

b) Show how your finding in part (a) can be used to show that $\sum_{n=0}^{\infty} \frac{5}{4 \cdot 8^n} = \frac{10}{7}$.